Jmar L. Pineda

Electrical and Computer Engineer

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PROFILE

Sophomore studying Electrical and Computer Engineering at NYIT with experience in hardware and software development, alongside undergraduate research. Project leader for the NYIT IEEE Micromouse team and programming instructor with a record of leading teams, mentoring students, and shipping low-level projects in C, C++, Rust, and Python.

EDUCATION

New York Institute of Technology — Sep 2025 – Dec 2029 (expected)
Electrical and Computer Engineering — New York, NY

- Cumulative GPA: 3.7

Queens High School for the Sciences — Sep 2021 – Jun 2025
High School Diploma — Jamaica, NY

- Cumulative GPA: 95.0

EXPERIENCE & LEADERSHIP

Open-Source Contributor — 2024 – 2026
Penrose & math.js — Remote

- **Penrose** (TypeScript, 4 merged PRs): eliminated editor memory leaks from detached SVG elements, mouse event listeners, and render state pileup during resample/compile (e.g. $\sim 46{,}000$ leaked listeners per compile down to 0), and wrote a benchmark script to profile leaks (#1961).
- **Penrose:** reused a single MathJax pipeline across compiles, cutting leaked heap by $\sim 85\%$ and leaked DOM nodes by $\sim 80\%$ (#1976); fixed an editor race condition on rapid recompiles (#1972).
- **math.js** (JavaScript): sped up the matrix `map()` and `forEach()` functions by over 300%, benefiting many of the library's other algorithms (#3251).

IEEE Micromouse Project Leader — Nov 2025 – Present
NYIT IEEE — New York, NY

- Lead and delegate tasks for a team of 5 to source materials, build the micromouse prototype and its software, create physical test cases, and iterate through rapid prototyping.
- Team is far ahead in progress compared to all other NYIT teams, driven by clear task ownership and collaboration.

Treasurer — Apr 2026 – Present
NYIT IEEE — New York, NY

- Manage the chapter's finances, budgeting, and forecasting.

Code Coach — Jan 2026 – Present
theCoderSchool — Syosset, NY

- Teach students aged 9–16 to make their first computer games, building a solid foundation in programming.
- Track the progress of 20+ students and design individualized lessons in Unity, Godot, Scratch, Roblox Studio, Arduino, Python, web development, VEX Robotics, Blender, and 3D printing.

Campus Events Coordinator — Nov 2025 – Present
NYIT Campus Activities Board — New York, NY

- Planned, hosted, and budgeted multiple campus-wide events in collaboration with faculty, clubs, and vendors.

Guest Coordinator & Volunteer — Mar 2017 – Present
Members Church of God International — New York, NY

- Coordinate with other guest coordinators to run events on schedule, welcome guests, maintain order, and serve food (Apr 2025 – Present).
- Volunteer at community service events including blood drives, food drives, clean-ups, fundraisers, and free grocery stores.

Peer Programming Tutor & Hackathon Leader Sep 2021 – Jun 2025
QHSS Hack Club Jamaica, NY

- Taught classmates to make games and websites, set appropriate challenges, and helped them through problems.
- Led my team to 1st place in the school-hosted Hackathon by delegating tasks and tackling the hardest problems.

Embedded Systems Programmer 2021 – 2025
QHSS Robotics Club Jamaica, NY

- Worked with builders and designers to define robot movements and goals, and optimized the robot for inter-school competitions.

PROJECTS

Triangle Slapper (C) github.com/Galm007/triangle_slapper

- Multi-threaded genetic algorithm that recreates an image from layered triangles: it spawns random triangles, scores and selects the best, and breeds the next generation.

Flappy Bird Neural Network (C) github.com/Galm007/flappy_bird_AI

- Neural network that learns to play Flappy Bird through a genetic algorithm, built on my Raylib game.

Forex Data Preprocessor (Python) github.com/Galm007/forex-data-preprocesor

- Computes common forex indicators from historical price and volume data, normalizes them, and packs them into a format ready for AI model training.

Metaballs Visualization (Rust, C#) macroquad | MonoGame

- Marching-squares metaballs implemented with macroquad (Rust) and MonoGame (C#), based on Jamie Wong's article.

Automated Forex Trading Algorithms (MQL4) github.com/Galm007/MT4-Expert-Advisors

- Series of trading algorithms for MetaTrader 4, with the most recent being the best performing.

SKILLS

Languages & Tools: C, C++, C#, Java, Python, Lua, JavaScript, Rust, HTML, CSS, SQL, Arduino, React, MQL4, Git, Docker, Linux, Shell, Excel

Areas: Embedded systems, robotics, game development (Unity, Godot, Roblox), AI implementations, trading algorithm development, web development

Soft skills: Communication, leadership, teamwork, project management

Languages spoken: Native English and Filipino; limited working proficiency in Spanish